

SMART STUDENT ID CARD MANAGEMENT PORTAL USING AWS

Prepared in the partial fulfillment of the Summer Internship Program on AWS

AT



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In conclusion, I am honoured to have been a part of this internship program, and I look forward to leveraging the skills and knowledge gained to contribute positively to future endeavours.

Thank you.

Sincerely,

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ABSTRACT

Cloud Based Smart Student ID Card Management System Using AWS

The Cloud Based Smart Student ID Card Management System is developed to simplify and automate the process of managing student identity cards in educational institutions. In the traditional system, students submit manual applications for new ID cards or lost ID card replacements, which is time-consuming, difficult to track, and requires a significant amount of paperwork. This project provides a secure and user-friendly web application that digitizes the entire process through cloud technology.

The system allows students to register, log in, apply for a new ID card, report a lost ID card, and track the status of their requests online. Staff members verify student requests, while administrators manage users, approve or reject requests, and oversee the complete system. All student and request information is stored securely in a centralized database, reducing manual effort and improving efficiency.

The application is developed using Java, JSP, Servlets, JDBC, Apache Tomcat, and MySQL. To provide cloud-based deployment and secure access, the project uses several Amazon Web Services (AWS) including Amazon EC2 for hosting the web application, Amazon RDS for database management, Amazon S3 for storing student photographs, Amazon VPC for secure networking, AWS Security Groups for controlling network access, and Amazon CloudWatch for monitoring the cloud resources.

The system offers improved security, centralized data management, faster processing, and easy accessibility through the internet. It minimizes paperwork, reduces processing time, and provides a reliable platform for managing student identity cards. The project demonstrates how cloud computing technologies can be effectively integrated with web application development to build a secure, scalable, and efficient student management solution.

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1. INTRODUCTION

The rapid growth of information technology and cloud computing has transformed the way educational institutions manage student information. Traditional methods of maintaining student identity card records involve manual paperwork, physical verification, and repeated visits to administrative offices. These methods are time-consuming, require more human effort, and increase the chances of errors such as misplaced records, duplicate entries, and delays in processing.

A Student Identity Card plays an important role in every educational institution. It serves as an official proof of identity for students and is used for library access, examination verification,

attendance monitoring, laboratory access, hostel management, and other academic activities. Managing student ID cards manually becomes increasingly difficult as the number of students grows every academic year.

To overcome these challenges, a Cloud Based Smart Student ID Card Management System has been developed. This project automates the complete process of student identity card management using cloud computing technologies. The system enables students to register online, log into the portal, apply for new ID cards, report lost ID cards, and track the status of their applications without visiting the college office multiple times.

The application is developed using Java Web Technologies, including JSP, Servlets, JDBC, HTML, and CSS. The project is deployed on Amazon Web Services (AWS) using Amazon EC2 as the application hosting server and Amazon RDS for cloud database management. Student photographs and related documents are stored securely using Amazon S3, while Amazon CloudWatch is used for monitoring cloud resources. The communication between cloud resources is protected using Amazon VPC and AWS Security Groups.

The proposed system provides improved security, centralized data management, faster processing, scalability, and easy accessibility through the Internet. It significantly reduces paperwork and manual effort while improving the efficiency of the student identity card management process.

1.2 Existing System

In the existing system, all student ID card-related activities are performed manually. Students are required to fill physical application forms and submit them to the college office. Staff members manually verify the details, prepare the ID card, and inform the student after completion.

This traditional approach has several disadvantages. Maintaining physical records consumes a large amount of storage space and increases the risk of losing important information. Students need to visit the administration office multiple times to know the application status. Staff members spend considerable time searching records and updating application details manually. There is no centralized system to manage student identity card information efficiently.

1.3 Proposed System

The proposed system introduces a cloud-based web application that automates the entire student ID card management process. Students can access the system from anywhere using an Internet connection. They can register, log in, apply for new ID cards, report lost ID cards, and track application status online.

Staff members verify submitted requests through their dashboard, while administrators approve or reject applications and manage the overall system. The application is deployed on Amazon EC2, while all data is securely stored in Amazon RDS. Student photographs are maintained in Amazon S3. CloudWatch continuously monitors system performance to ensure smooth operation.

1.4 Objectives

The major objectives of the project are:

- To automate the student ID card management process.
- To eliminate manual paperwork.
- To provide secure cloud-based data storage.
- To allow students to submit applications online.
- To enable online tracking of application status.
- To improve administrative efficiency.
- To provide secure communication between application and database.
- To reduce processing time.
- To increase scalability using AWS cloud services.

1.5 Advantages

The proposed system offers several advantages:

- Paperless process
- Faster request processing
- Secure cloud storage
- Easy accessibility
- Centralized database management
- Better monitoring
- Reduced human effort
- Improved security
- Better scalability
- Cost-effective maintenance

2. METHODOLOGY

The development of the Cloud Based Smart Student ID Card Management System follows a systematic software development methodology. The project begins with requirement analysis, where the problems of the traditional system are identified and user requirements are collected.

The application is designed using Java Web Technologies. Eclipse IDE is used as the development environment. User interfaces are created using HTML, CSS, and JSP pages. Business logic is implemented using Java Servlets. Database connectivity is established through JDBC.

Initially, the application is developed and tested on the local machine. After successful testing, the project is exported as a WAR (Web Application Archive) file. The WAR file contains all project resources, including JSP pages, Servlets, configuration files, and libraries.

The exported WAR file is uploaded to an Amazon EC2 instance running Apache Tomcat. Apache Tomcat acts as the web server that executes the Java web application. The application communicates with Amazon RDS through JDBC for storing and retrieving student information.

Student photographs and related documents are stored securely using Amazon S3. CloudWatch monitors EC2 and RDS performance, including CPU utilization, memory usage, and network activity. Amazon VPC provides an isolated virtual network for secure deployment, while Security Groups control inbound and outbound traffic.

The methodology ensures that the application is secure, scalable, reliable, and easily accessible through the Internet.

3. TECHNOLOGY USED

Java

Java is an object-oriented programming language widely used for developing enterprise-level web applications. Java provides platform independence through the Java Virtual Machine (JVM), allowing applications to run on different operating systems without modification.

Java Servlets are used in this project to implement the business logic. Servlets receive client requests, process them, communicate with the database, and generate dynamic responses

JSP (Java Server Pages)

Java Server Pages are used to develop dynamic web pages. JSP simplifies the development of user interfaces by embedding Java code within HTML pages. In this project, JSP pages are used to create login, registration, student dashboard, staff dashboard, and administrator dashboard.

JDBC

Java Database Connectivity (JDBC) is an API that establishes communication between Java applications and relational databases. JDBC enables the application to perform CRUD operations such as inserting, updating, deleting, and retrieving records from Amazon RDS.

Apache Tomcat

Apache Tomcat is an open-source web server and Servlet container. It executes JSP pages and Servlets and handles client requests. The project is deployed on Tomcat running inside Amazon EC2.

MySQL

MySQL is a relational database management system used to store student information, login details, ID card requests, and lost ID card requests. Amazon RDS provides a managed MySQL database service in the cloud.

AWS Services

Amazon EC2

Amazon EC2 hosts the Java web application and provides computing resources.

Amazon RDS

Amazon RDS securely stores application data and manages the MySQL database.

Amazon S3

Amazon S3 stores student photographs and documents

Amazon CloudWatch

CloudWatch continuously monitors cloud resources and application performance.

Amazon VPC

Amazon VPC provides a secure virtual private network where EC2 and RDS communicate safely.

AWS Security Groups

Security Groups function as virtual firewalls, controlling inbound and outbound network traffic between cloud resources.

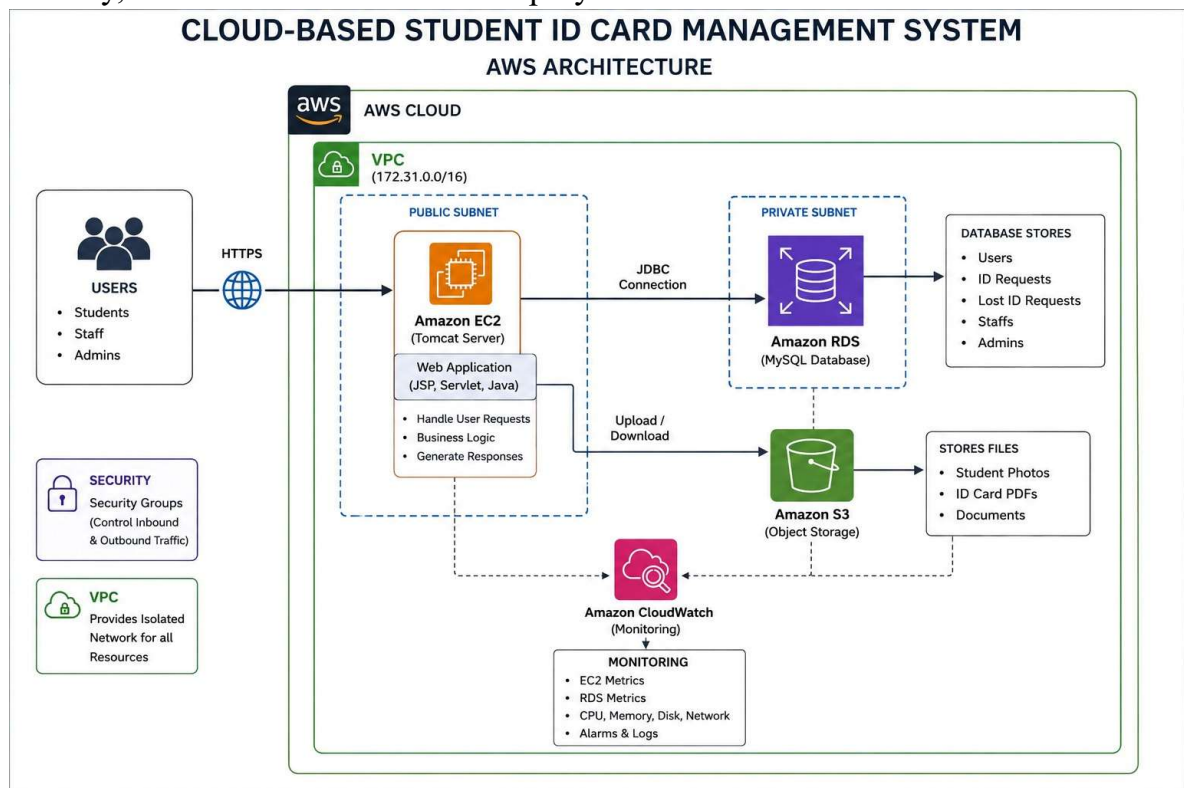
4. SYSTEM DESIGN / ARCHITECTURE

The Cloud Based Smart Student ID Card Management System follows a three-tier cloud architecture consisting of the Presentation Layer, Application Layer, and Database Layer. The application is deployed on Amazon Web Services (AWS) to provide secure, scalable, and reliable access over the Internet.

Students, staff members, and administrators access the system through a web browser using the public IP address of the Amazon EC2 instance. User requests such as login, registration, ID card applications, and lost ID requests are first received by the Apache Tomcat web server running on the EC2 instance. The Java Servlets process these requests, execute the business logic, and communicate with the Amazon RDS MySQL database using JDBC.

Whenever student photographs are uploaded, they are stored in Amazon S3 instead of the local server. This ensures reliable cloud storage with high durability and availability. Amazon CloudWatch continuously monitors the health and performance of EC2 and RDS resources, including CPU utilization, memory usage, and network traffic. Amazon VPC provides an isolated private network for AWS resources, while AWS Security Groups act as virtual firewalls to protect the application from unauthorized access.

The architecture ensures high availability, centralized management, improved security, and efficient cloud-based deployment.



Architecture Components

1. Client Layer

The client layer consists of students, staff members, and administrators who access the application through a web browser using the public IP address of the Amazon EC2 instance.

Users can perform the following operations:

Student Registration

Login

Apply for ID Card

Lost ID Request

Track Status

Staff Verification

Admin Approval

2. Application Layer

The application layer is hosted inside Amazon EC2.

The EC2 instance runs Apache Tomcat, which hosts the Java web application.

Main responsibilities include:

Receiving HTTP Requests

Executing JSP Pages

Running Java Servlets

Business Logic Processing

Database Communication

Apache Tomcat acts as the servlet container that processes all incoming requests and generates dynamic responses.

3. Business Logic Layer

The business logic is implemented using Java Servlets.

Each Servlet performs a specific task.

Examples:

LoginServlet

RegisterServlet

RequestServlet

LostIdServlet

ApproveServlet

RejectServlet

The Servlets validate user inputs, execute SQL queries, interact with Amazon RDS, and return appropriate responses.

4. Database Layer

The database layer is implemented using Amazon RDS MySQL.

Amazon RDS stores:

User Information

Student Details

Login Credentials

ID Card Requests

Lost ID Requests

Request Status

Advantages of using Amazon RDS include:

Automatic Backup

High Availability

managed Database Service

Easy Scalability

Secure Storage

5. Storage Layer

Student photographs are stored in Amazon S3.

Instead of storing images inside the web server, Amazon S3 provides secure cloud storage.

Benefits include:

High Durability

High Availability

Easy Image Management

Scalable Storage

Low Maintenance

6. Monitoring Layer

amazon CloudWatch continuously monitors cloud resources.

It collects:

CPU Utilization

Memory Usage

Network Traffic

Storage Metrics

Logs

CloudWatch helps administrators identify performance issues and maintain system reliability.

7. Security Layer

Security is provided using:

Amazon VPC

Amazon VPC creates a secure private network where EC2 and RDS communicate.

Security Groups

Security Groups act as virtual firewalls.

They allow only authorized traffic.

examples:

Port 8080 → Web Application

Port 22 → SSH

Port 3306 → MySQL Database

Unauthorized users cannot directly access the database.

4.4 System Workflow

The working process of the system is as follows:

User opens the application using the EC2 public IP.

The request reaches Apache Tomcat.

Tomcat forwards the request to the appropriate Java Servlet.

The Servlet processes the request.

JDBC establishes a connection with Amazon RDS.

Data is inserted, updated, or retrieved.

If photographs are uploaded, they are stored in Amazon S3.

CloudWatch monitors the performance of EC2 and RDS.

The processed response is returned to the user's browser.

Advantages of the Architecture

Cloud-Based Deployment

High Availability

Secure Database Access

Centralized Data Management

Image Storage using Amazon S3

Real-Time Monitoring

Easy Scalability

Improved Security

Better Performance Low Maintenance Cost

5. IMPLEMENTATION

5.1 introduction

The implementation phase is one of the most important stages of the project development process. During this phase, the designed system is converted into a working web application. The Cloud Based Smart Student ID Card Management System was developed using Java Web Technologies and successfully deployed on Amazon Web Services (AWS).

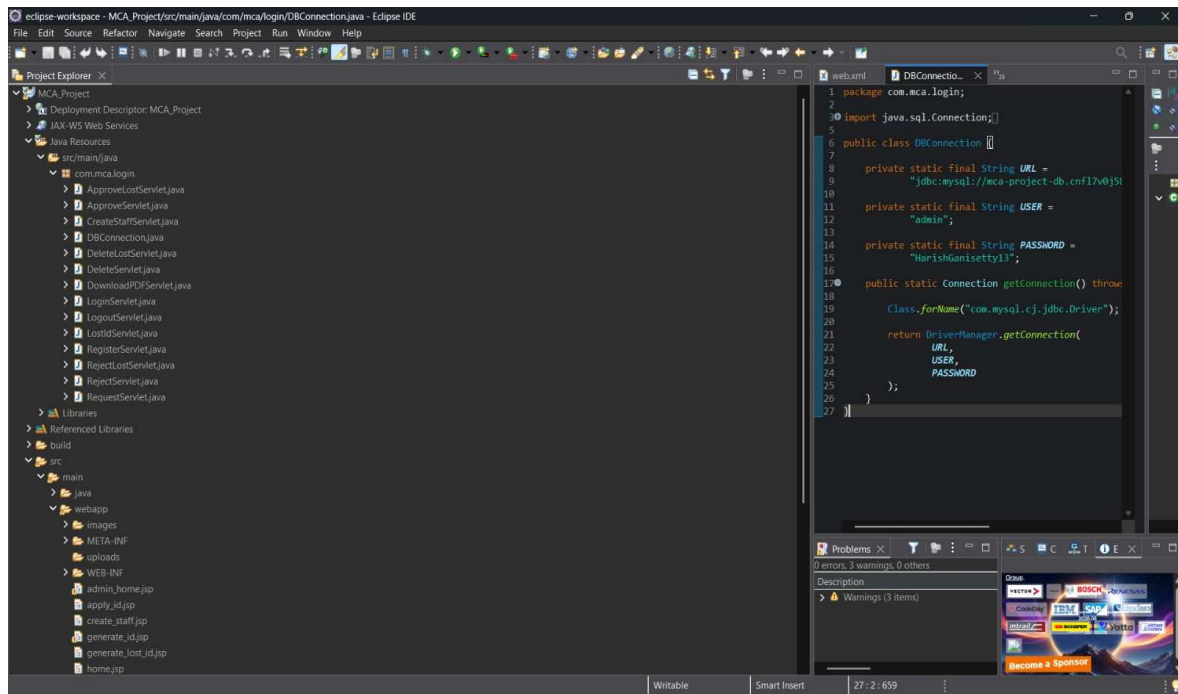
The application was first developed and tested in the Eclipse IDE using Java, JSP, Servlets, JDBC, HTML, and CSS. After successful testing, the project was exported as a WAR (Web Application Archive) file and deployed on Apache Tomcat running inside an Amazon EC2 instance. Amazon RDS was used for cloud database management, while Amazon S3 was configured for storing student photographs. Amazon CloudWatch was used to monitor the performance of cloud resources.

5.2 development environment

The project was developed using the following software and cloud services:

Component	Technology Used
IDE	Eclipse IDE
Programming Language	Java
Frontend	HTML, CSS, JSP
Backend	Java Servlets
Database	MySQL
Database Connectivity	JDBC
Web Server	Apache Tomcat
Cloud Platform	Amazon Web Services
Cloud Database	Amazon RDS

Storage	Amazon S3
Monitoring	Amazon CloudWatch
Networking	Amazon VPC
Security	AWS Security Groups



5.3 Database Implementation

Initially, the database was created in MySQL. The required tables were created to store user information, ID card requests, and lost ID card requests.

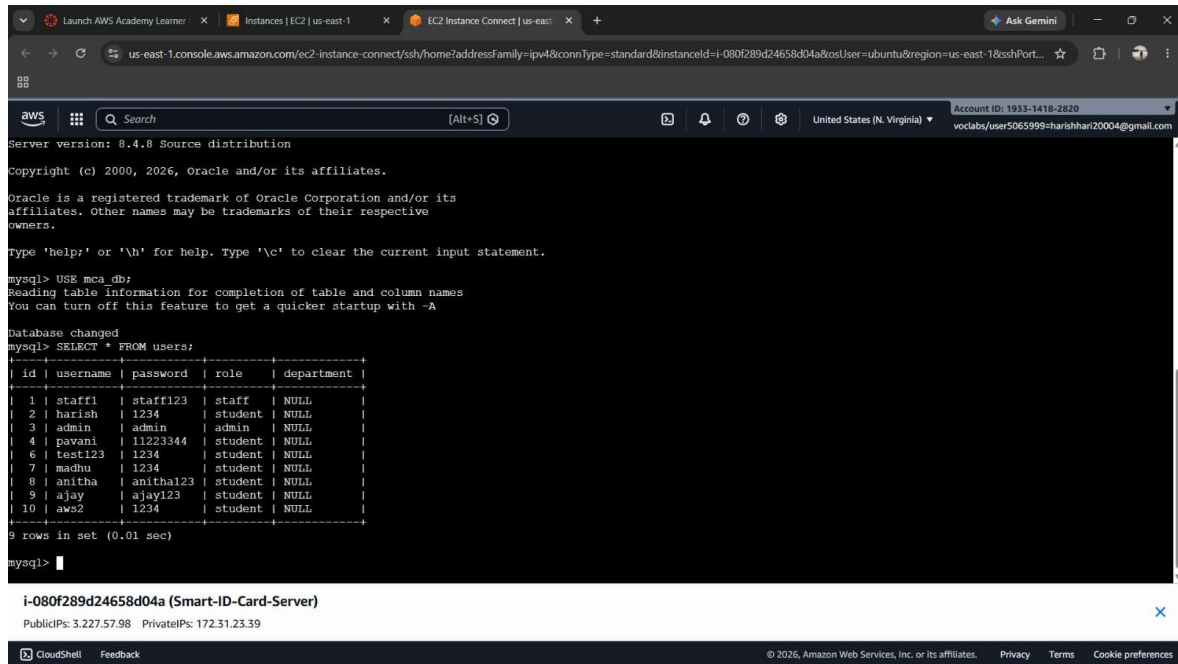
The main database tables are:

users

id_requests

lost_id_requests

The database was later migrated to Amazon RDS to enable cloud-based data storage.



The screenshot shows an AWS CloudShell terminal window. The terminal output displays the MySQL server version (8.4.8) and the source distribution. It then shows the execution of a query to select all data from the 'users' table. The results are displayed in a table format with columns: id, username, password, role, and department. The table contains 10 rows of data, including staff, students, and an admin user. The terminal also shows the prompt 'mysql>' and the command 'USE mca_db;'.

```
Server version: 8.4.8 Source distribution
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE mca_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

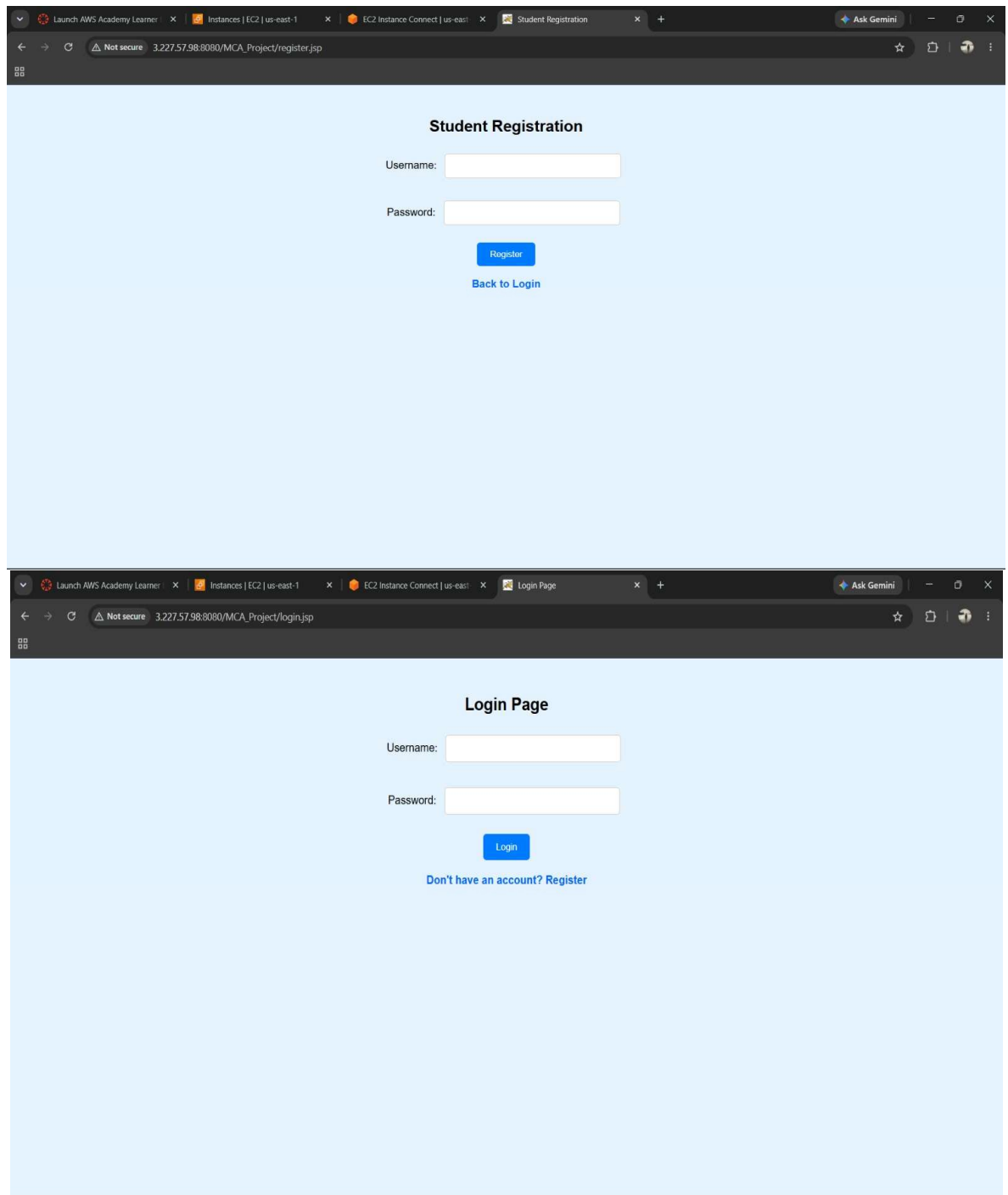
Database changed
mysql> SELECT * FROM users;
+----+-----+-----+-----+-----+
| id | username | password | role | department |
+----+-----+-----+-----+-----+
| 1 | staff1 | staff123 | staff | NULL |
| 2 | harish | 1234 | student | NULL |
| 3 | admin | admin | admin | NULL |
| 4 | pavani | 11223344 | student | NULL |
| 6 | test123 | 1234 | student | NULL |
| 7 | madhu | 1234 | student | NULL |
| 8 | anitha | anitha123 | student | NULL |
| 9 | ajay | ajay123 | student | NULL |
| 10 | aws2 | 1234 | student | NULL |
+----+-----+-----+-----+-----+
9 rows in set (0.01 sec)

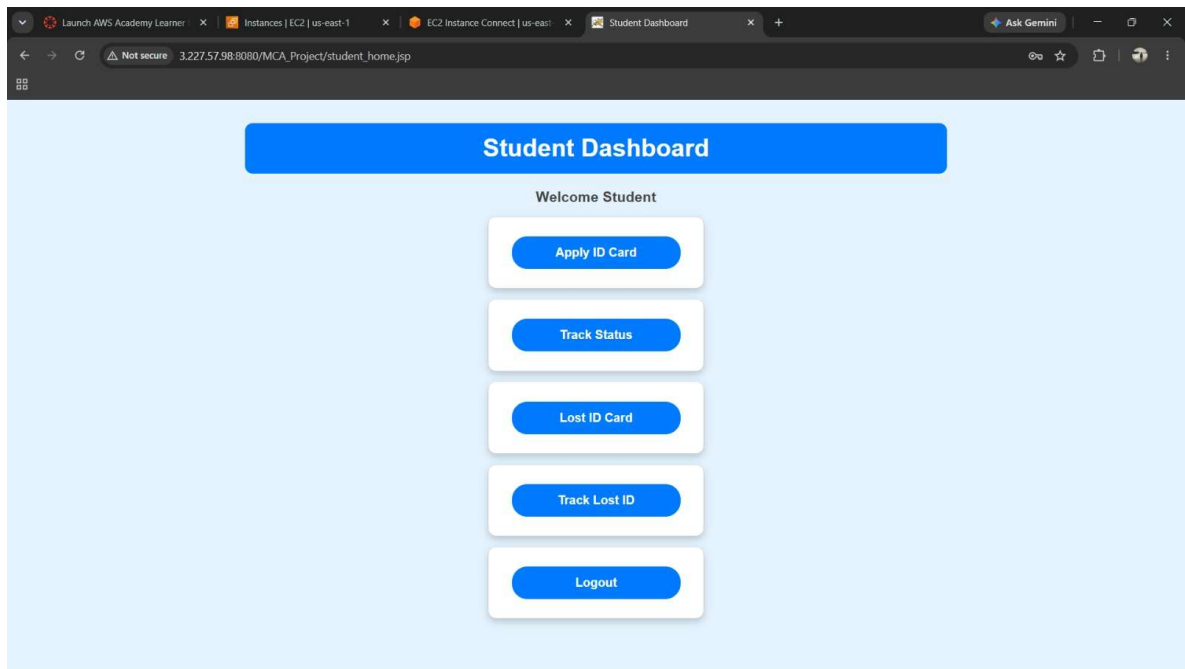
mysql>
```

Application Development

The application consists of three different user modules.

- Student Module
- Registration
- Login
- Apply ID Card
- Lost ID Card
- Track Status





Staff Module

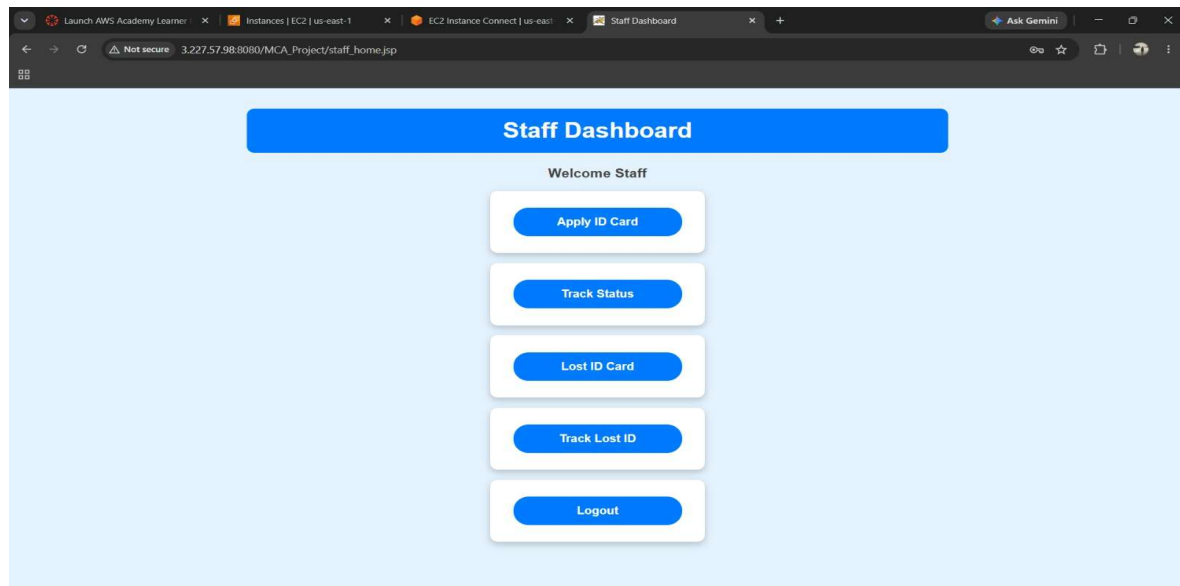
The Staff module allows staff members to:

Verify Requests

View Student Requests

Approve Requests

Reject Requests



Admin Module

The administrator manages the complete application.

Features include:

Create Staff

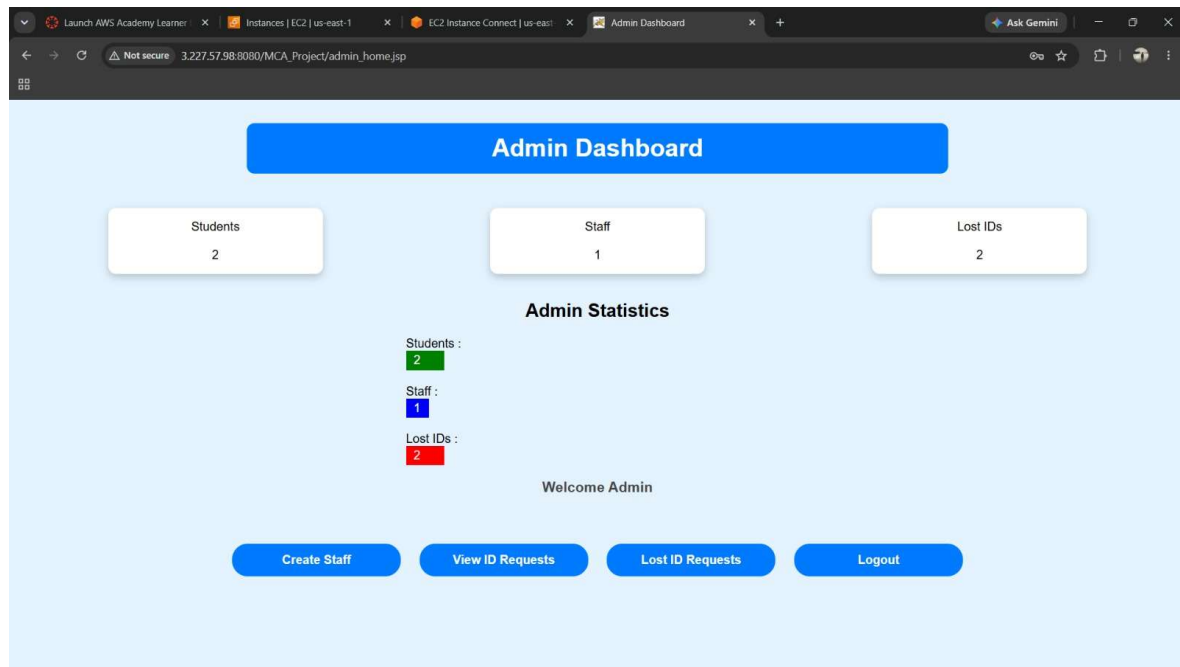
View Students

View Requests

Approve Requests

Reject Requests

Delete Requests



5.5 AWS Cloud Deployment

After successful testing, the project was exported as a WAR file.

The WAR file was uploaded into Amazon EC2 using SCP.

Apache Tomcat was installed inside Ubuntu running on Amazon EC2.

WAR file was deployed inside the Tomcat webapps folder.

Tomcat automatically extracted the project and hosted the application.

Application URL:

http://3.227.57.98:8080/MCA_Project

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page is active, displaying a list of EC2 instances. One instance, 'Smart-ID-Card-Server' (ID: i-080f289d24658d04a), is highlighted. Its details are shown below the list, including its 'Running' status, 't3.micro' type, and various network addresses. The left sidebar shows navigation options like Dashboard, Events, and various instance management tools.

UBUNTU TERMINAL

This screenshot shows a terminal session within the AWS CloudShell environment. The terminal displays system information as of Monday, July 13, 2026, at 10:00:18 UTC. It lists system load, memory usage, and network configuration, including the public IP address 3.227.57.98 and private IP address 172.31.23.39. The terminal also shows security update information and the current directory path ~/tomcat/webapps.

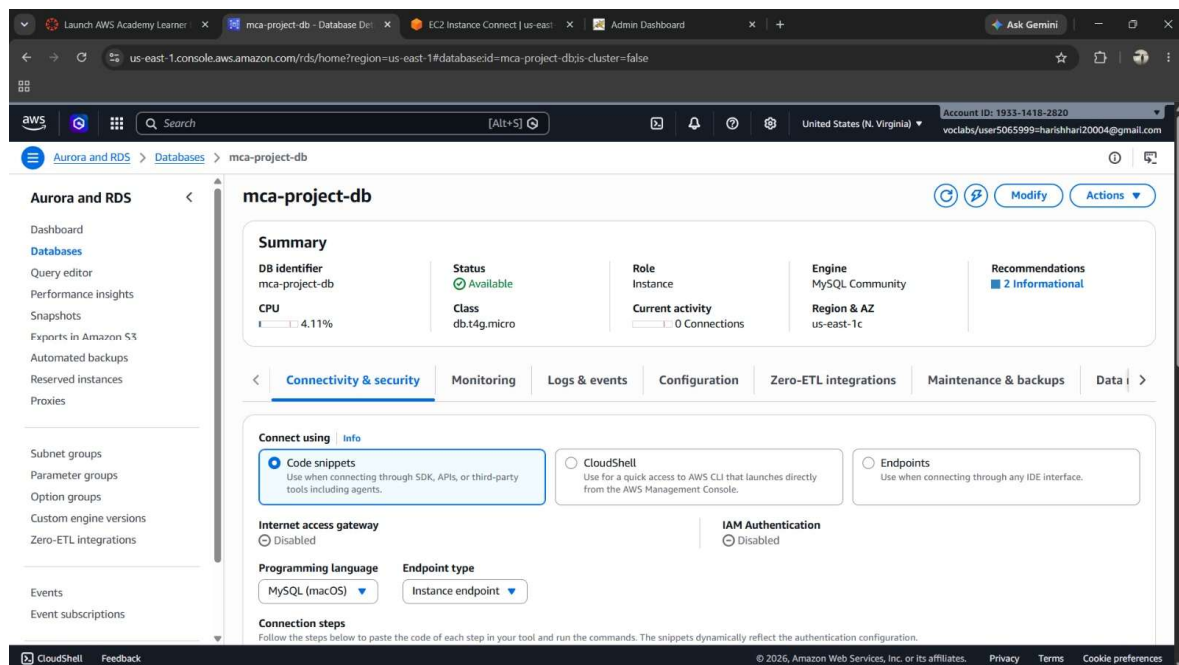
5.6 Amazon RDS Integration

The local MySQL database was replaced with Amazon RDS.

The DBConnection.java file was modified to connect with Amazon RDS.

JDBC Driver established communication between Java application and Amazon RDS.

Student data is stored directly in the cloud database.



5.7 Amazon S3 Implementation

Amazon S3 was used for cloud object storage.

Student photographs can be uploaded and stored inside an S3 Bucket.

The bucket created

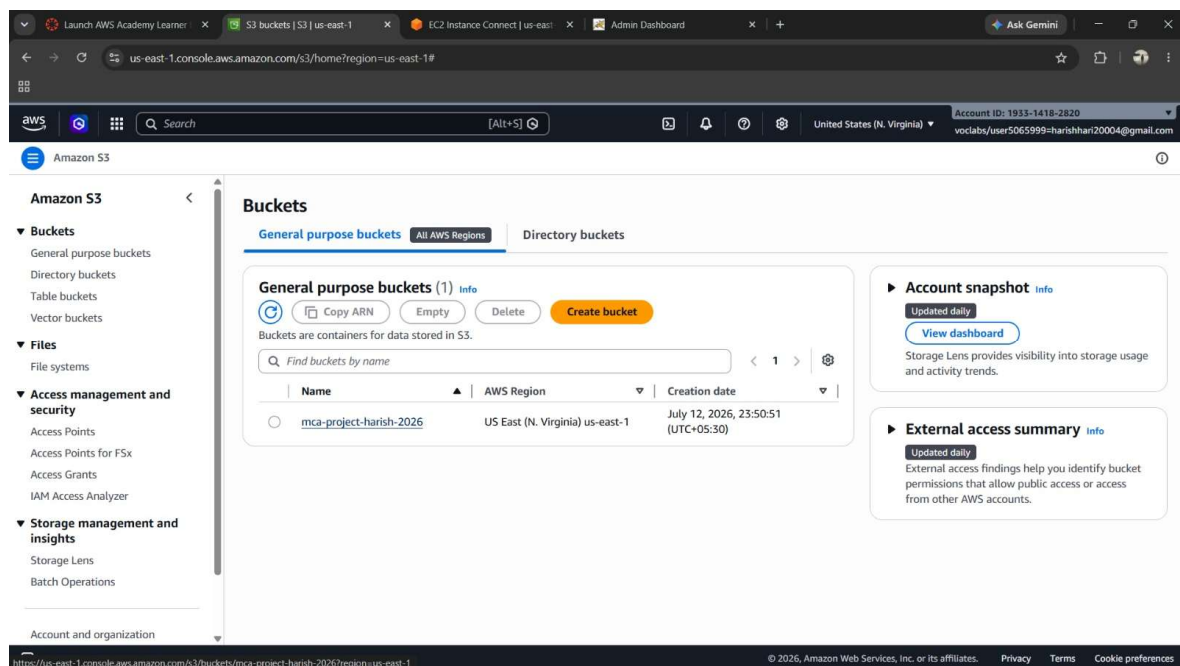
mca-project-harish-2026

Benefits:

Secure Storage

High Availability

Easy File Management



5.8 Amazon CloudWatch

Amazon CloudWatch continuously monitors the application.

CloudWatch collects:

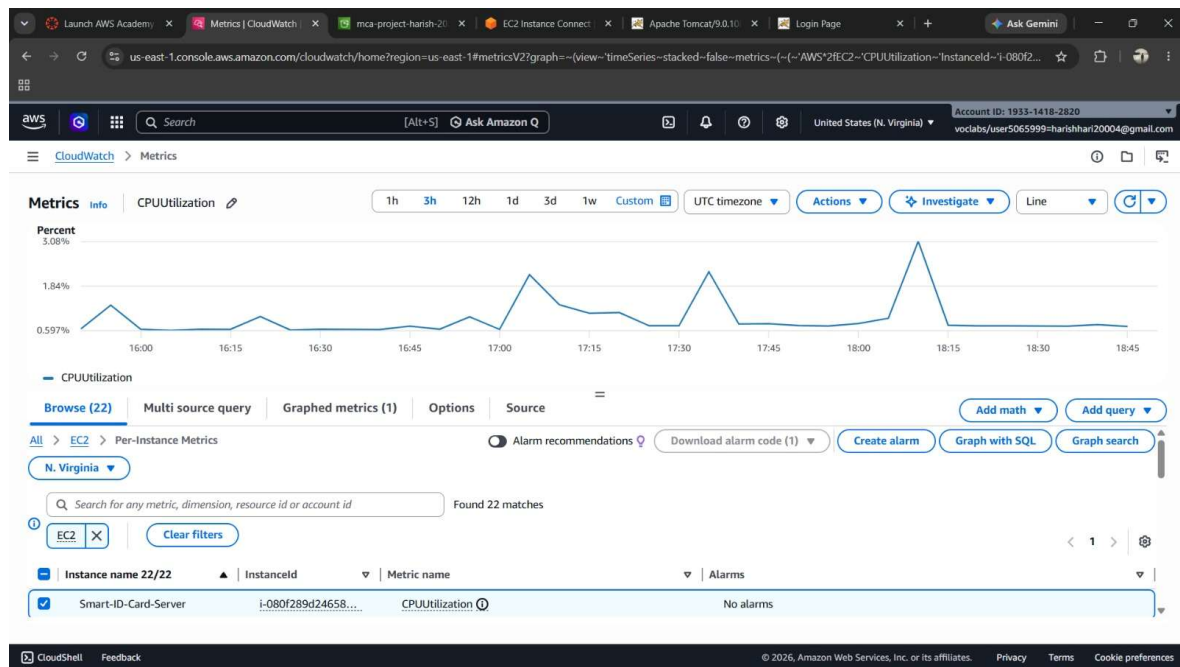
CPU Utilization

Network Traffic

Disk Usage

Instance Status

CloudWatch helps administrators monitor application health



5.8 Amazon CloudWatch

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CloudWatch collects

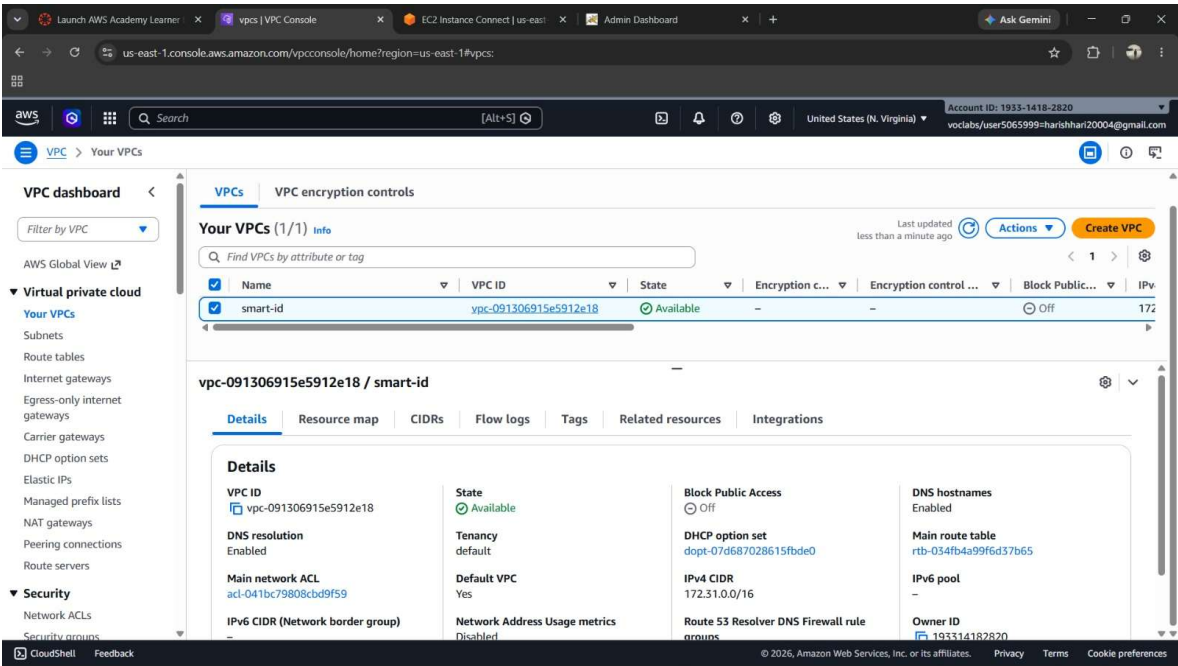
CPU Utilization

Network Traffic

Disk Usage

Instance Status

CloudWatch helps administrators monitor application health



5.11 Testing

The application was tested successfully.

Testing included:

Student Registration

Login

Apply ID

Lost ID

Track Status

Staff Approval

Admin Approval

Database Update

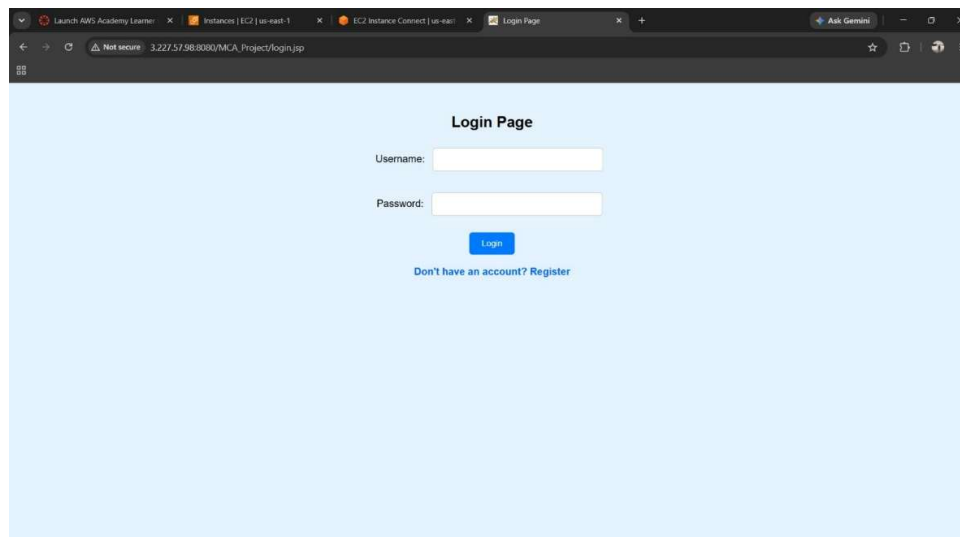
After testing, new users were successfully inserted into Amazon RDS.

6. RESULTS

6.1 Introduction

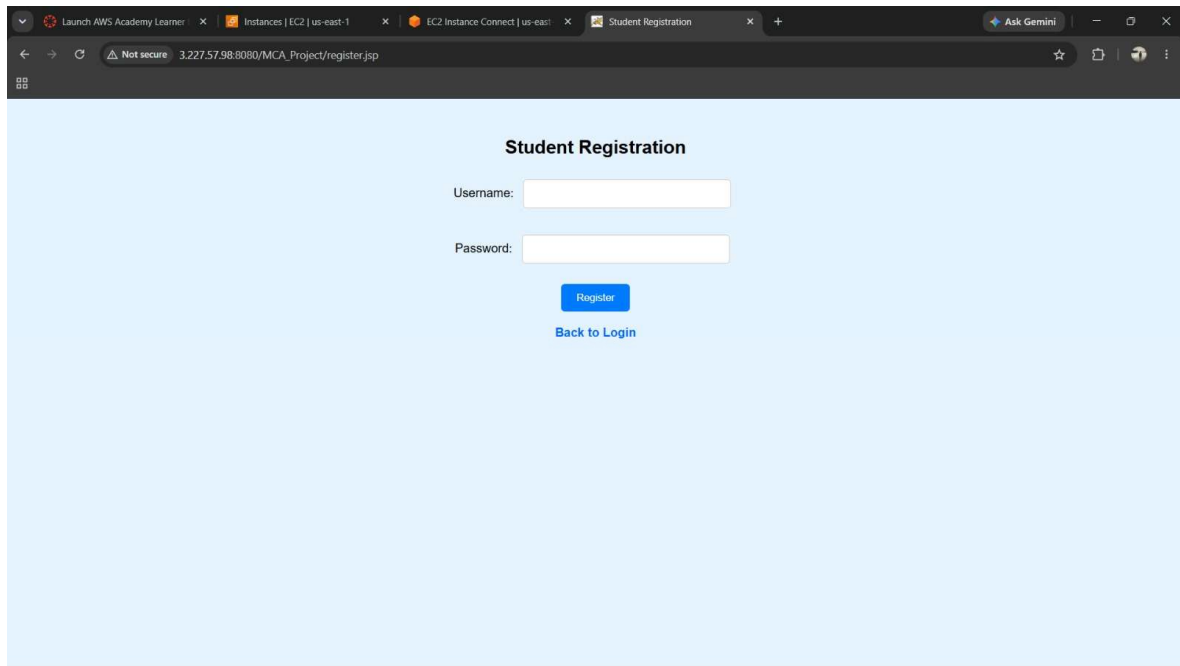
The Cloud Based Smart Student ID Card Management System was successfully developed, tested, and deployed on Amazon Web Services (AWS). The application was implemented using Java Web Technologies and hosted on Amazon EC2 with Apache Tomcat. Amazon RDS was used as the cloud database, Amazon S3 for cloud storage, Amazon CloudWatch for monitoring, and Amazon VPC with Security Groups for secure networking. All the modules were tested successfully and produced the expected results.

LOGIN PAGE



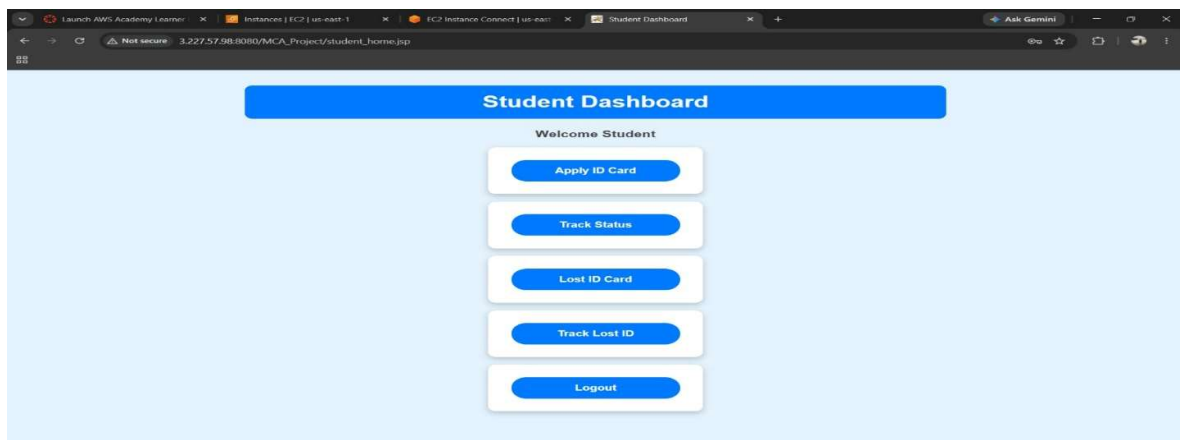
The login page is the entry point of the application. It allows students, staff members, and administrators to log into the system using their registered username and password. Based on the user's role, the system redirects them to their respective dashboard after successful authentication.

REGISTRATION PAGE

A screenshot of a web browser showing the 'Student Registration' page. The browser's address bar displays 'Not secure 3.227.57.98.8080/MCA_Project/register.jsp'. The page has a light blue background. At the top center, the title 'Student Registration' is displayed. Below the title, there are two input fields: 'Username:' and 'Password:'. A blue 'Register' button is positioned below the password field, and a blue 'Back to Login' link is located directly beneath the 'Register' button.

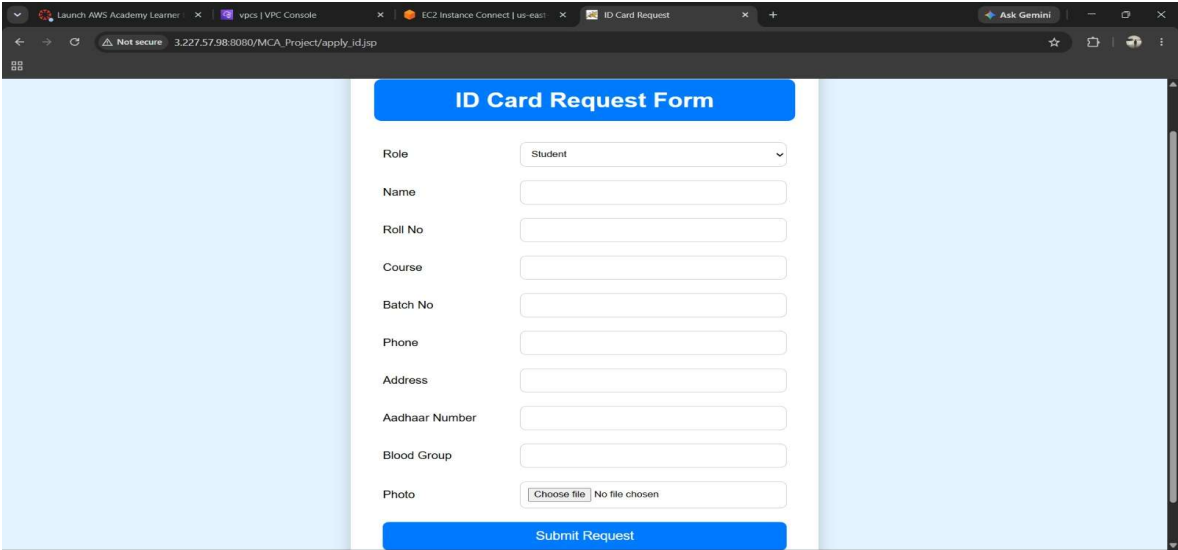
The registration page allows new students to create an account by entering their username and password. The application checks for duplicate usernames before inserting the new record into the Amazon RDS database. After successful registration, the user is redirected to the login page.

STUDENT DASHBOARD

A screenshot of a web browser showing the 'Student Dashboard' page. The browser's address bar displays 'Not secure 3.227.57.98.8080/MCA_Project/student_home.jsp'. The page has a light blue background. At the top center, there is a blue header bar with the text 'Student Dashboard'. Below this bar, the text 'Welcome Student' is displayed. Underneath, there is a vertical stack of five blue buttons with white text: 'Apply ID Card', 'Track Status', 'Lost ID Card', 'Track Lost ID', and 'Logout'.

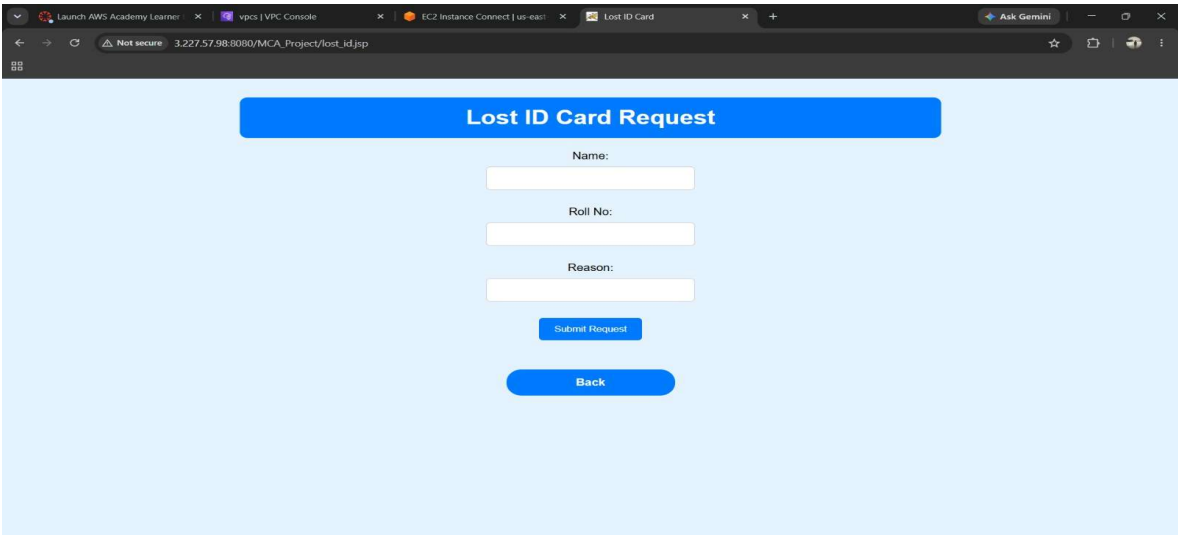
After successful login, students are redirected to the Student Dashboard. From this dashboard, students can apply for a new ID card, report a lost ID card, and track the status of their submitted requests. This module provides a simple and user-friendly interface.

APPLY ID CARD PAGE



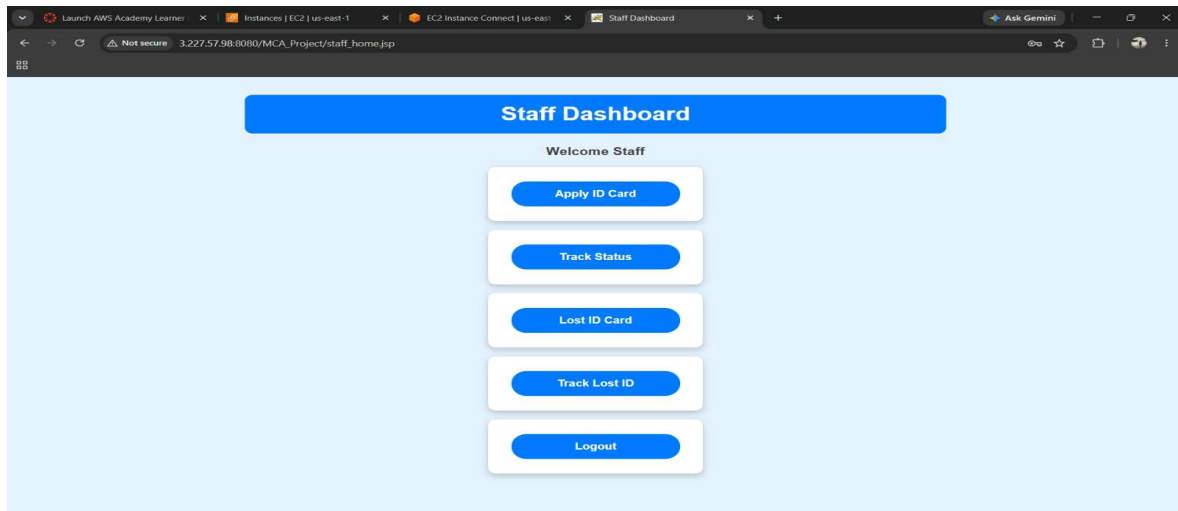
The Apply ID Card page allows students to submit a request for a new identity card by providing the required details. The submitted information is stored securely in the Amazon RDS database and is available for staff verification and administrator approval.

LOST ID CARD PAGE



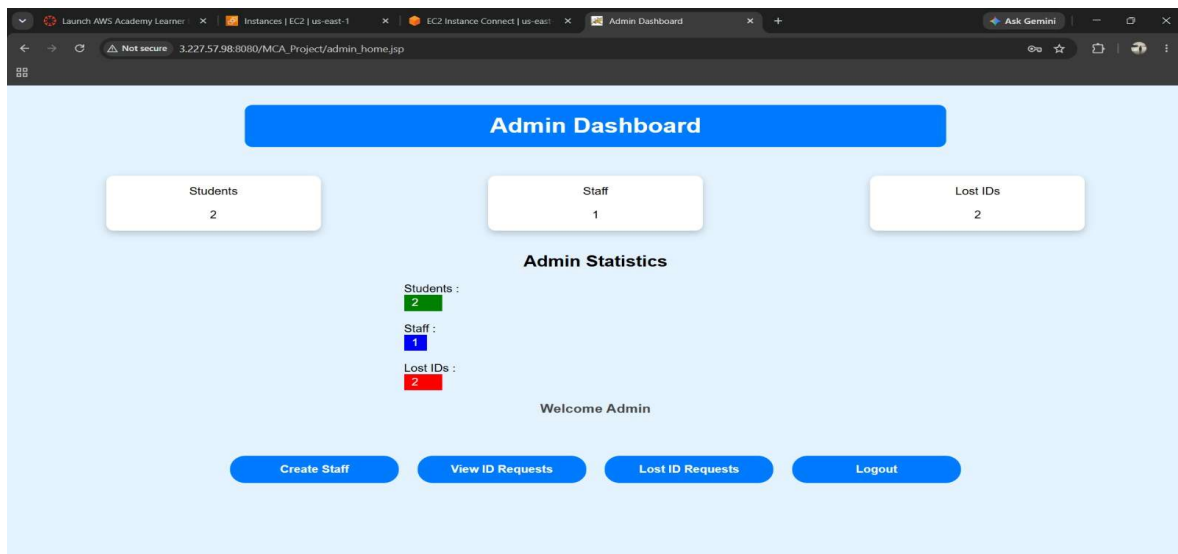
The Lost ID Card page enables students to report a lost identity card. The application stores the request in the database and forwards it to the staff and administrator for further processing.

STAFF DASHBOARD



The Staff Dashboard provides authorized staff members with access to student ID card requests. Staff members can review applications, verify student details, and forward valid requests for administrator approval.

ADMIN DASHBOARD



The Administrator Dashboard provides complete control over the application. The administrator can manage users, create staff accounts, approve or reject student requests, monitor system activities, and manage all records efficiently.

7.CONCLUSION

The Cloud Based Smart Student ID Card Management System successfully automates the traditional manual process of student identity card management. The project demonstrates how cloud computing technologies can be integrated with Java web applications to provide secure, scalable, and reliable solutions.

The application allows students to register, log in, apply for new ID cards, report lost ID cards, and track application status through an online portal. Staff members verify applications while administrators manage the overall system efficiently.

The deployment of the application on Amazon EC2, integration with Amazon RDS, storage using Amazon S3, monitoring through Amazon CloudWatch, and secure networking using Amazon VPC and Security Groups demonstrate the practical implementation of cloud computing technologies.

The proposed system reduces paperwork, minimizes processing time, improves data security, enhances administrative efficiency, and provides better accessibility to users. It can be extended further by integrating QR code generation, biometric authentication, mobile applications, and AI-based verification techniques.